

# Lecture 1

## From the shape Hessian to its eigenvalues

Detailed explanations accompanying the torsion manuscript

13 September 2026

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# 1 The question and the coordinates

**Aim.** Starting from Laurain's distributed second shape derivative, obtain the two eigenvalues of every Fourier block of the vertex Hessian. A suggested 90-minute session spends 20 minutes on vertex calculus, 25 on the local blocks, 30 on Fourier reduction, and 15 on the checks and exercises. The prerequisite is the weak Dirichlet problem and the meaning of a second shape derivative; no interval arithmetic is used in this lecture.

For a polygon  $P$ , let

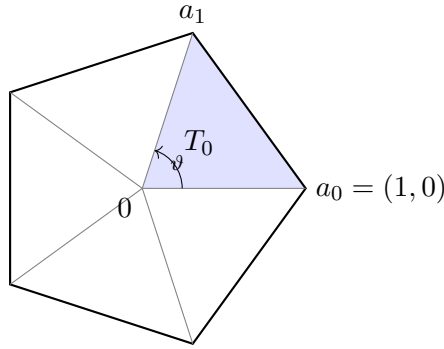
$$-\Delta u = 1 \quad \text{in } P, \quad u \in H_0^1(P), \quad J(P) = \frac{1}{2} \int_P u = \frac{1}{2} \int_P |\nabla u|^2.$$

Our objective is  $\mathcal{F}(P) = J(P)/A(P)^2$ , where  $A(P) = |P|$ . Conventional torsional rigidity for this PDE is  $2J$ . Multiplying the objective by two changes eigenvalue magnitudes, but not their signs. The dilation laws  $u_{tP}(tx) = t^2 u_P(x)$  and  $J(tP) = t^4 J(P)$  explain the power  $A^2$  in the denominator.

Fix the regular polygon of circumradius one. Throughout the course,

$$\vartheta = \frac{2\pi}{n}, \quad a_j = (\cos(j\vartheta), \sin(j\vartheta)), \quad T_j = \text{conv}\{0, a_j, a_{j+1}\}.$$

Indices are taken modulo  $n$ . The same  $\vartheta$  is used everywhere; we do not give separate names to its half-angle, its multiples, or its sine and cosine. The rotation through  $j\vartheta$  is  $R_j$ . The vector of vertex coordinates is interleaved as  $(x_0, y_0, x_1, y_1, \dots)$ .



Let  $\phi_j$  be the continuous affine hat on this coarse fan, one at  $a_j$  and zero at its other vertices, including the centre. A vertex perturbation  $q = (q_j)_{j=0}^{n-1}$ ,  $q_j \in \mathbb{R}^2$ , is extended into the polygon by

$$\theta_q(x) = \sum_j \phi_j(x) q_j.$$

This formula says exactly how points inside the polygon move: the map  $x \mapsto x + s\theta_q(x)$  moves the boundary vertices as prescribed and fixes the centre. For sufficiently small  $s$ , it is a bi-Lipschitz map. Its gradient is constant on each fan triangle. This last fact will be used in every subsequent calculation.

## 2 From a shape derivative to a vertex Hessian

For each vertex,  $U_i = (U_i^1, U_i^2)$  collects the material derivatives of  $u$  under displacements of that vertex in the two coordinate directions. The superscript is a component, not a derivative order. Differentiating the pulled-back weak equation gives, for scalar  $v \in H_0^1(P)$ ,

$$\begin{aligned} \int_P DU_i \nabla v &= \int_P -(\nabla u \cdot \nabla v) \nabla \phi_i + (\nabla \phi_i \cdot \nabla v) \nabla u \\ &\quad + (\nabla \phi_i \cdot \nabla u) \nabla v + v \nabla \phi_i. \end{aligned} \tag{1}$$

For the direction  $q$ , the scalar material derivative is  $u_q = \sum_i q_i \cdot U_i$ . All these functions have zero Dirichlet trace on the reference polygon: the boundary condition was pulled back before being differentiated.

We assume Laurain's distributed formula [2, Proposition 14]. Its planar reduction is manuscript equation 8:

$$\begin{aligned} (D^2 J)_{ij} &= \int_P DU_i DU_j^T \\ &\quad + \int_P \left( \frac{1}{2} |\nabla u|^2 + u \right) (\nabla \phi_i \otimes \nabla \phi_j - \nabla \phi_j \otimes \nabla \phi_i) \\ &\quad - \int_P (\nabla \phi_i \cdot \nabla \phi_j) (\nabla u \otimes \nabla u). \end{aligned} \quad (2)$$

This is a  $2 \times 2$  block. For example its Gram contribution in component  $(p, q)$  is  $\int_P \nabla U_i^p \cdot \nabla U_j^q$ .

Here is the algebra behind the reduction. For any three planar vectors  $p, r, g$ ,

$$\begin{aligned} (r \cdot g)p \otimes g + (p \cdot g)g \otimes r - (p \cdot g)r \otimes g - (r \cdot g)g \otimes p \\ = |g|^2(p \otimes r - r \otimes p). \end{aligned}$$

Checking the two diagonal entries gives zero on both sides; checking one off-diagonal entry gives the other by antisymmetry. Substitute  $p = \nabla \phi_i$ ,  $r = \nabla \phi_j$ ,  $g = \nabla u$  in Laurain's integrand. The remaining terms proportional to  $u$  give the skew expression in (2). This is a two-dimensional simplification; it should not be used unchanged in dimension three.

There are two useful structural checks. First, individual blocks need not be symmetric, but  $(D^2 J)_{ji} = ((D^2 J)_{ij})^T$ . Second, the last two terms are local: the hats have overlapping gradients only for identical or neighboring vertices. The Gram term is the only generally dense part.

## 2.1 The area correction must be differentiated too

The shoelace formula gives

$$A = \frac{1}{2} \sum_j (x_j y_{j+1} - y_j x_{j+1}), \quad \nabla_{a_i} A = \frac{1}{2} (y_{i+1} - y_{i-1}, x_{i-1} - x_{i+1}).$$

The only nonzero mixed second derivatives connect neighboring vertices:  $A_{x_i y_{i+1}} = 1/2$  and  $A_{y_i x_{i+1}} = -1/2$ , together with their symmetric counterparts. At the regular polygon,  $A = n \sin \vartheta / 2$  and  $\nabla_{a_i} A = \sin \vartheta a_i$ .

By the ordinary product rule,

$$\begin{aligned} D^2 \mathcal{F} &= A^{-2} D^2 J - 2JA^{-3} D^2 A \\ &\quad - 2A^{-3} (\nabla J \otimes \nabla A + \nabla A \otimes \nabla J) + 6JA^{-4} \nabla A \otimes \nabla A. \end{aligned} \quad (3)$$

Dihedral symmetry makes  $\nabla_{a_i} J$  a common multiple of  $a_i$ . Euler's identity for the degree-four functional  $J$  then gives

$$\nabla J = \frac{2J}{A} \nabla A, \quad D\mathcal{F}(P_n) = 0.$$

Substitution into (3) yields

$$D^2 \mathcal{F} = A^{-2} D^2 J - 2JA^{-3} D^2 A - 2JA^{-4} \nabla A \otimes \nabla A. \quad (4)$$

This formula is valid at the regular polygon. On the area-tangent space,  $D^2 J - (2J/A) D^2 A$  has the same signs as  $D^2 \mathcal{F}$ . One must not inspect  $D^2 J$  alone: radial dilation increases  $J$ .

### 3 Compute the local part on two triangles

The hat gradients at vertex zero are

$$\nabla\phi_0 = (1, -\cot\vartheta) \text{ on } T_0, \quad \nabla\phi_0 = (1, \cot\vartheta) \text{ on } T_{-1},$$

and zero elsewhere. Also,  $\nabla\phi_1 = (0, 1/\sin\vartheta)$  on  $T_0$  and  $\nabla\phi_{-1} = (0, -1/\sin\vartheta)$  on  $T_{-1}$ . These identities follow by solving the three nodal interpolation equations, not by differentiating a trigonometric interpolant.

Only three sector integrals are needed:

$$X = \int_{T_0} u_x^2, \quad Y = \int_{T_0} u_y^2, \quad Z = \int_{T_0} u_x u_y.$$

Reflection across each radial side makes  $\partial_n u = 0$  there. On the polygon side  $u = 0$ . Integration by parts within  $T_0$  therefore gives

$$X + Y = \int_{T_0} u = \frac{2J}{n}, \quad \int_{T_0} (\tfrac{1}{2}|\nabla u|^2 + u) = \frac{3J}{n}.$$

Reflection across the sector bisector diagonalizes the sector energy tensor in the bisector basis. In Cartesian coordinates this means

$$(Y - X) \sin\vartheta + 2Z \cos\vartheta = 0. \tag{5}$$

In particular,  $-X \sin\vartheta + Z \cos\vartheta = -J \sin\vartheta/n$ . This identity is the cancellation to remember; storing  $Z$  separately is unnecessary for the final symbol.

Write  $L_j$  for the first block row of the local, non-Gram part of  $D^2J$ . Substitution of the hat gradients into (2) gives

$$\begin{aligned} L_0 &= -\frac{2}{\sin^2\vartheta} \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}, \\ L_1 &= \frac{\cos\vartheta}{\sin^2\vartheta} \begin{pmatrix} X & Z \\ Z & Y \end{pmatrix} - \frac{3J}{n \sin\vartheta} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \\ L_{-1} &= \frac{\cos\vartheta}{\sin^2\vartheta} \begin{pmatrix} X & -Z \\ -Z & Y \end{pmatrix} + \frac{3J}{n \sin\vartheta} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \end{aligned} \tag{6}$$

All other  $L_j$  vanish. To check the sign in  $L_1$ , note that  $\nabla\phi_0 \cdot \nabla\phi_1 = -\cos\vartheta/\sin^2\vartheta$ , whereas  $\nabla\phi_0 \otimes \nabla\phi_1 - \nabla\phi_1 \otimes \nabla\phi_0$  is the negative of the displayed rotation matrix divided by  $\sin\vartheta$ .

#### 3.1 What remains unknown in the sector integrals?

If  $\Delta_n$  denotes the difference of the energies parallel and perpendicular to the sector bisector, rotation of that diagonal tensor gives

$$X = \frac{J}{n} + \frac{\Delta_n}{2} \cos\vartheta, \quad Y = \frac{J}{n} - \frac{\Delta_n}{2} \cos\vartheta, \quad Z = \frac{\Delta_n}{2} \sin\vartheta.$$

Symmetry alone does not determine  $\Delta_n$ . The manuscript's mixed Dirichlet–Neumann torsion problem provides an analytic characterization of it; the fixed-mesh certificate does not need that separate computation. For the square,  $X = Y = J/4$  even though  $Z$  need not vanish. The sector is not symmetric under reflection in the Cartesian axes individually.

## 4 Why radial and tangential coordinates reveal the spectrum

At vertex  $j$ , the radial unit vector is  $a_j$  and the tangential unit vector is  $(-\sin(j\vartheta), \cos(j\vartheta))$ . They are the columns of  $R_j$ . With  $R = \text{diag}(R_0, \dots, R_{n-1})$ , set

$$H = R^T D^2 \mathcal{F} R.$$

This is an orthogonal change of coordinates, so eigenvalues are unchanged. In these coordinates rotation of the polygon merely shifts the vertex index. Consequently  $H_{ij} = C_{j-i}$ , where  $C_{-j} = C_j^T$ .

The block-circulant reduction is due to Tee [3] and is used in this polygonal setting by Bogosel and Bucur [1]. It can be proved in one line. For a sequence  $v_j = e^{ikj\vartheta} w$ ,  $w \in \mathbb{C}^2$ ,

$$(Hv)_i = \sum_j C_{j-i} e^{ikj\vartheta} w = e^{iki\vartheta} \left( \sum_{j=0}^{n-1} e^{ikj\vartheta} C_j \right) w.$$

Thus the  $k$ th Fourier subspace carries the  $2 \times 2$  matrix

$$B_k = \sum_{j=0}^{n-1} e^{ikj\vartheta} C_j. \quad (7)$$

The normalized discrete Fourier transform is unitary; the multiset of all  $2n$  Hessian eigenvalues is the union of the spectra of these  $n$  blocks.

Symmetry gives  $B_k^* = B_k$ . Reflection in the axis through vertex zero acts as  $\text{diag}(1, -1)$  on radial/tangential components and sends  $j$  to  $-j$ . The diagonal sequences are even and the off-diagonal sequence is odd. Therefore

$$B_k = \begin{pmatrix} \alpha_k & i\gamma_k \\ -i\gamma_k & \beta_k \end{pmatrix}, \quad \mu_k^\pm = \frac{\alpha_k + \beta_k \pm \sqrt{(\alpha_k - \beta_k)^2 + 4\gamma_k^2}}{2}. \quad (8)$$

The parameters  $\alpha_k, \beta_k, \gamma_k$  are real. These are manuscript equations (39)–(40). The sign of the imaginary entry depends on the Fourier convention; the convention in (7) is fixed throughout these notes.

### 4.1 The explicit cancellation with area

The radial/tangential area symbol is

$$\sin \vartheta \cos(k\vartheta) I - i \cos \vartheta \sin(k\vartheta) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In the first row, the local blocks become  $L_j R_j$ , since  $R_0 = I$ . Multiply the three matrices in (6) by  $R_j$ , use (5), and subtract the area contribution in (4). The result is

$$\begin{aligned} A^{-2} \sum_{j=-1}^1 e^{ikj\vartheta} L_j R_j - 2JA^{-3} \left[ \sin \vartheta \cos(k\vartheta) I - i \cos \vartheta \sin(k\vartheta) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] \\ = -\frac{2(1 - \cos(k\vartheta))}{A^2 \sin^2 \vartheta} \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}. \end{aligned} \quad (9)$$

The explicit off-diagonal term disappears completely. The rank-one term  $\nabla A \otimes \nabla A$  in (4) acts only on the constant radial sequence, hence only on mode zero.

For clarity, introduce the scalar rotated components already used in the manuscript:

$$U_{j,r} = \cos(j\vartheta)U_j^1 + \sin(j\vartheta)U_j^2, \quad U_{j,t} = -\sin(j\vartheta)U_j^1 + \cos(j\vartheta)U_j^2, \quad a(v, w) = \int_P \nabla v \cdot \nabla w.$$

The three genuinely nonlocal quantities are

$$\begin{aligned} G_k^{rr} &= \sum_j \cos(kj\vartheta) a(U_0^1, U_{j,r}), \\ G_k^{tt} &= \sum_j \cos(kj\vartheta) a(U_0^2, U_{j,t}), \\ G_k^{rt} &= \sum_j \sin(kj\vartheta) a(U_0^1, U_{j,t}). \end{aligned}$$

For every nonzero mode, (9) now yields

$$\boxed{\begin{aligned} \alpha_k &= A^{-2} \left[ G_k^{rr} - \frac{2(1 - \cos(k\vartheta))X}{\sin^2 \vartheta} \right], \\ \beta_k &= A^{-2} \left[ G_k^{tt} - \frac{2(1 - \cos(k\vartheta))Y}{\sin^2 \vartheta} \right], \\ \gamma_k &= A^{-2} G_k^{rt}. \end{aligned}} \quad (10)$$

These are the requested Hessian eigenvalue formulas together with (8). Their trace contains the explicit penalty  $-4(1 - \cos(k\vartheta))J/(nA^2 \sin^2 \vartheta)$ . Sector anisotropy cancels from that explicit penalty; it does not cancel from the state-dependent Gram terms.

## 5 Exact zero modes and the local maximum criterion

At a critical point of a similarity-invariant function, differentiating invariance puts the four infinitesimal similarity directions in the Hessian kernel. The mode bookkeeping matters:

| motion          | radial/tangential coordinates           | Fourier location |
|-----------------|-----------------------------------------|------------------|
| scaling         | (1, 0) at every vertex                  | $k = 0$          |
| rotation        | (0, 1) at every vertex                  | $k = 0$          |
| $x$ translation | $(\cos(j\vartheta), -\sin(j\vartheta))$ | $k = 1, n - 1$   |
| $y$ translation | $(\sin(j\vartheta), \cos(j\vartheta))$  | $k = 1, n - 1$   |

Thus  $B_0 = 0$ . At  $k = 1$  the complex translation vector is  $(1, i)$ , so  $\alpha_1 = \beta_1 = \gamma_1$ . The other eigenvalue is the trace  $\alpha_1 + \beta_1$ . The conjugate mode  $n - 1$  gives the other real translation and the same nonzero eigenvalue.

Also  $B_{n-k} = \overline{B_k}$ . If  $n$  is even, the Nyquist mode  $k = n/2$  has zero sine sequence and  $\gamma_{n/2} = 0$ . These are exact identities, not quantities to be recognized by a small numerical threshold. For a generic block, strict negativity is equivalent to

$$\alpha_k + \beta_k < 0, \quad \alpha_k \beta_k - \gamma_k^2 > 0.$$

This determinant test excludes translation blocks, whose determinant is exactly zero; there the trace alone is tested.

**Theorem 5.1** (What the numerical calculation must establish). *If every branch outside the four similarity directions is strictly negative, the regular polygon is a strict local maximizer of  $J/A^2$  modulo similarities. Equivalently, it is a strict local maximizer of  $J$  at fixed area, modulo translations and rotations.*

*Proof.* The fan pullback has smoothly parameter-dependent, uniformly elliptic coefficients in a sufficiently small vertex neighborhood. Its inverse solution operator is smooth, so the objective is twice differentiable. Choose a local slice complementary to the four-dimensional similarity orbit. The Hessian is negative definite on the slice; Taylor's theorem gives strict decrease for small nonzero slice displacements. Every nearby orbit meets the slice. Fixing area removes the dilation freedom. This proves a local statement about nearby labelled polygons, not global optimality among all polygons.  $\square$

## 6 The quantities actually passed to the next lecture

One need not assemble the entire Hessian to evaluate its symbol. For  $1 \leq k < n/2$ , the following are normalized vertex displacements:

$$\begin{aligned} r_{k,j}^c &= \sqrt{2/n} \cos(kj\vartheta) (\cos(j\vartheta), \sin(j\vartheta)), & \text{radial, cosine,} \\ t_{k,j}^c &= \sqrt{2/n} \cos(kj\vartheta) (-\sin(j\vartheta), \cos(j\vartheta)), & \text{tangential, cosine,} \\ t_{k,j}^s &= \sqrt{2/n} \sin(kj\vartheta) (-\sin(j\vartheta), \cos(j\vartheta)), & \text{tangential, sine.} \end{aligned}$$

Their pairings give respectively

$$D^2\mathcal{F}[r_k^c, r_k^c] = \alpha_k, \quad D^2\mathcal{F}[t_k^c, t_k^c] = \beta_k, \quad D^2\mathcal{F}[r_k^c, t_k^c] = 0, \quad D^2\mathcal{F}[r_k^c, t_k^s] = \gamma_k.$$

For the Nyquist cosine directions use  $(-1)^j/\sqrt{n}$  instead of  $\sqrt{2/n} \cos(kj\vartheta)$ . The certificate encloses these real pairings directly; it need not compute separate intervals for  $X, Y, Z$  or the Gram sums.

Finally, covariance gives

$$U_j(x) = R_j U_0(R_j^T x), \quad U_{j,r}(x) = U_0^1(R_j^T x), \quad U_{j,t}(x) = U_0^2(R_j^T x).$$

The distinction between vector rotation and scalar pullback explains the optimized code: torsion and two first-vertex PDEs supply every first material derivative. Later second-lifting solves serve error estimation.

### Exercises and answers for discussion

**Exercise 6.1.** *Derive  $\nabla J = (2J/A)\nabla A$  without a boundary integral formula.*

*Answer.* Reflection makes each gradient radial; rotation makes their coefficients equal. If  $\nabla_{a_j} J = ca_j$ , Euler's identity gives  $nc = 4J$ . Compare with  $\nabla_{a_j} A = \sin \vartheta a_j$ .

**Exercise 6.2.** *For  $n = 5$ , explain why the target is six negative eigenvalues, even though there are only two representative nonzero modes.*

*Answer.* Modes 1, 4 contribute one physical eigenvalue each; modes 2, 3 contribute two each. Mode zero and the two translations supply four zeros. The total is  $2 + 4 = 6$  negatives out of ten dimensions.

**Exercise 6.3.** *Can a negative computed trace alone certify a generic  $2 \times 2$  block?*

*Answer.* No. A block with eigenvalues  $-2$  and  $1$  has negative trace. A determinant bound or two strictly negative eigenvalue enclosures is also required.

## References

- [1] B. Bogosel and D. Bucur. On the polygonal Faber–Krahn inequality. *Journal de l'École polytechnique—Mathématiques*, 11:19–105, 2024.
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